Assignment 4: Relational Databases vs NoSQL

Library Management System – MySQL

Introduction

A relational database stores structured information in tables with rows and columns. NoSQL databases support flexible data models such as documents, key-value pairs, wide columns and graphs.

Relational database approach

A library system is well suited to a relational database because books, members, loans and fines have clear relationships. SQL supports joins, constraints and transactions. Examples include MySQL, PostgreSQL and Oracle Database.

NoSQL approach

A document database can store flexible records, for example a book document containing title, publication details, tags and embedded review data. Examples include MongoDB, Cassandra and Redis (a key-value store). NoSQL can be useful for rapidly changing or very large distributed data.

Comparison

Relational databases generally provide strong schema structure, joins and transactional consistency. NoSQL systems often provide flexible schemas and convenient horizontal scaling, depending on the product and configuration. Neither model is universally better; the choice depends on data structure and application requirements.

Recommendation for the Library Management System

MySQL is a suitable choice because the system needs reliable relationships between books, members, loans, reservations and fines. Foreign keys help prevent invalid references, and transactions can ensure that borrowing and fine updates are handled consistently.

Conclusion

For this library schema, a relational DBMS is the most practical choice. NoSQL could be added later for a separate feature such as activity logs or flexible recommendation metadata.